A Gapless Waveform Recorder for Monitoring Smart Grids

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***Abstract*—In the present paper a new concept of continuous waveform recording, or gapless waveform recording, applied to Electric Power System (EPS) signals is presented. It is shown that it is not necessary to store all the EPS signal samples, and still, keeping all the relevant information for a complete reconstruction of the signal for later oscillographic analysis. This allows store long periods of waveform measurements, spending little memory space. To make it possible, it is used the Novelty Detection Concept, that is performed by segmenting the signal into frames and comparing these frames in order to save only the ones in which some "novelty" is present. This approach contributes to the new context of Smart Grids (SGs), even in a future scenario where events that are not completely known might occur, since all the necessary waveform information is saved. Thus, the control system of the SG can use the stored signals to perform different analysis and take the suitable decision to improve the EPS performance.**

***Index Terms*—Gapless Waveform Recording, Smart Grids, Novelty Detection, Power Quality.**

I. Introduction

The growing demand for electricity has been, evidently, one of the main reasons for the expansion of Electric Power Systems (EPSs). Along the recent years, the EPSs have become more and more ramified, feeding multiple types of loads, which drain utility power from the grid in increasingly peculiar form. Besides, the grid has become a place where non-linear and non-predictable loads are being constantly coupled and uncoupled [1], which increases the Power Quality (PQ) pollution caused by nonlinear characteristics of them [2].

Today, the utilities across the world are facing many challenges that were never before anticipated or prepared for [3]. Indeed, power system has become more and more sophisticated and drawn attention among power professionals and researchers along the years [4]–[7].

Because of the significant impact on the operation and economic consequences for many different types of facilities, Power Quality (PQ) issues have become a great concern [8]. Moreover, in the future, other electrical disturbances, not fully known may occur, and the way how they impact on PQ is a very important aspect for a good performance and proper operation of the system. Therefore, the system must be prepared for such events.

It is essential to improve the power grid so as it is suitable to the current expansion scenario, through the insertion of several technologies inside it, in order to transform it into a Smart Grid (SG). Key aspects of SGs are the detection, classification and identification of these disturbances [9]. To do this, digital signal and statistical processing techniques have been widely used to analyze power system signals and automatically detect and recognize PQ disturbances [10]–[13].

Besides, in several EPS applications such as PQ, operation and protection, continuous data acquisition at a high sampling rate and data storage are necessary [14]. In SG applications, devices that are able to accomplish such a task are even more needed, in order to identify other kinds of events.

A high potential use of such data is in the equipment condition monitoring, since many equipment failures present an unique signatures in the voltage and current waveforms. For example, high frequency disturbances like transient oscillations or switching processes, as the arcing of cable joint, restrike of a capacitor switch, and tree-contact by a power line are only visible in the full waveform, in which their signatures are present. Furthermore, applications involving finding signatures hidden in raw data can be used to predict events like breaking of cables, etc. [15]. Therefore, the storage of the full waveform is essential for the next generation of EPS, in the context of SG.

In order to deal with SG scenario possibly not fully known, new disturbances detection methods need to be developed, allied to the possibility of total reconstruction of the gapless waveform.

This paper aims to present a new approach to record the waveform of EPS signals by detecting novelties in the power signal, and storing them, so that a full reconstruction of the gapless waveform is possible, allowing later oscilographic analysis.

The paper is organized as follows. Section II discusses about the current state of PQ analyzers and monitors, presenting the need of gapless waveform recorders. Section III presents the

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Novelty Detection Concept applied to EPS waveform signals recording, where are present an explanation about this concept and a real example of application. Finally, Section IV states some conclusions about the topic addressed in the paper.

II. Current PQ Analyzers and Monitors

Nowadays, it is easy to find PQ recorders or data loggers that can save data from the power grid. But, in general only average values, calculated within aggregation intervals sometimes configurable, are stored. Although aggregation is useful for data reduction and comparison, and compliance to standards, deep data analysis should be enabled considering full observation possibilities, mainly in SG applications [16]. While agreement to these standards is useful for compliance investigations it is not necessarily useful for power system diagnostic purposes [17]–[19].

The term "Gapless Recording" have been used with double meaning in the technical literature. Sometimes, and more often, it is related to gapless PQ parameters, which means no losing of PQ parameters. In other cases, it is used as a continuous waveform recorder [20]. Nevertheless, in the new context of PQ in SGs it is possible that new events, not before observed, may occur. For this reason, it is important to avoid losing valuable information from the power signal.

A. Big-Data

One way to achieve the goal of keeping all the relevant information, is storing all the samples (raw data) [21]. Obviously, that will generate a lot of data. Even with data reduction provided by the use of aggregation values and triggered waveform recordings, the amount of PQ data is extremely large and it has aroused interest in studies around this problem. The current Digital Fault Recorders (DFRs) are facing this fact [22]. Considering the storage of all samples, these problems becomes even more critical.

Modern PQ monitors are based on computer technology, which has grown up at an incredible rate. Furthermore, while the cost of memory has decreased exponentially, the processing power of computers has increased at a similar rate in recent years. These advancements in computer technology have given rise to the development of "Big Data" and they have directly influenced modern PQ instrument technology [23].

Despite these facts, the memory space optimization has proven positive factor in tasks that require PQ data processing [24]. A device which can deal with an optimal storing usage is better than other that was designed without the concerning about memory space.

B. Examples of commercial Gapless Data Recorders

In order to show the current state of commercially available devices, a thorough survey of the leading manufacturers of PQ analyzers and data loggers was made. Nine of the top brands that describe their products as "Power Quality Analyzer", "Gapless Recorder", "Disturbance Recorder" or "Continuous Recorder" were examined. A total of 27 devices were analyzed, but, for sake of simplicity, only the most highly rated of each brand is presented. All the features of interest were extracted from the investigation of their manuals and datasheets, and the main data is summarized in Table I. Data not provided by the manufacturer are filled with "—" (without quotes). For ethical reasons, manufacturers are represented by capital letters.

In Table I, the second and third columns are related to continuous waveform recording capacity of each device. The second one reveals what is the sampling rate used by each equipment. Some equipment have a fixed sampling rate, a simple setting switches between 50 or 60 Hz systems. Others have a dynamic range of sampling rate. They use techniques like Phase Locked Loop (PLL) or Zero Crossing [25] in order to track the frequency of the input signal and to perform a synchronous sampling. The third column shows the waveform recording lengths of each equipment.

The fourth and fifth columns are related to data recording or parameter recording. Some equipment have several possibilities for choosing an interval for aggregation calculating, from some seconds up to hours. In some cases, the value of the interval is given in Hertz, which corresponds the frequency that the value is either calculated or taken from he waveform signal. In the fourth column the total period of parameter recording and the memory usage are presented. The last column has the memory capacity of each device.

From Table I the following remarks can be highlighted:

A - Sampling frequency with automatic fundamental frequency synchronization from 45 Hz to 65 Hz. Transient analysis (TA) sampling rate from 200 kHz to 2 MHz. While the maximum recording length at 2 MHz is 32 ms, at 200 kHz it is 320 ms. For continuous parameter recording it can be parameterized an interval from 1 second to 30 minutes. Internal memory of 4 GB permits several months of recording, but it can be extended to several years by an external SD card up to 32 GB.

B - Sampling frequency of 384 Samples per Cycle (S/C), which corresponds to 19.2/23.04 kHz for 50/60 Hz systems, respectively. It has a maximum recording length of 60 s in this fault recording mode for TA. For disturbances recording, it has a frequency of twice the supply frequency, that is, 100/120 Hz for 50/60 Hz systems, respectively. The maximum record length in this mode is 30 minutes. Another mode, commonly referred to as "Trend Recording", has either one or ten minutes of sampling interval. The maximum recording length in this mode is 52 weeks. The mass storage uses a 40 GB Solid State Drive (SSD), which can store over 1000 faults and disturbance records simultaneously.

C - Sampling frequency from 64 up to 1024 S/C, regardless of the supply frequency (50, 60 or 400 Hz). It stores the RMS data at the setted intervals which can be adjusted from 0.2 to 30 seconds or 1 to 30 minutes. The total record length using only this feature is not informed by the manufacturer, since it is done together with the continuous waveform recording. Using an SD card with 128 GB and setting the sampling rate to 128 S/C for

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TABLE I: Main features of best PQ analysers and data loggers currently commercially

Brand	Continuous Waveform Recording		Parameter Recording		Memory/ Mass Storage
	Sampling rate (50/60 Hz)	Record Lengths	Frequency (50/60 Hz) or Measurement interval	Memory usage or Recording lengths	
A	41 kHz @ 50 Hz 200 kHz - 2 MHz	320 ms (200 kHz) 32 ms (2 MHz);	Free interval 1 s - 1800 s (default: 10 min)	10 min int. $\rightarrow$ 15 MB/week 1 s int. $\rightarrow$ 15 MB$/\frac{1}{2}$ h	4 GB (extensible up to 32 GB, requires an SD card)
B	19.2 kHz @ 50 Hz 23.04 kHz @ 60 Hz	1 s - 60 s	100/120 Hz; 25/30 Hz; 1 min interval	30 min; 2 weeks; 52 weeks	40 GB (SSD)
C	64 - 1024 S/C (x 50/60 Hz)	with 128 S/C and 8 channel: 5 days (64 GB) 11 days (128 GB)	0.2 s - 30 s 1 min - 30 min interval	—	4 - 128 GB (SD card)
D	512 S/C (x 50/60 Hz) up to 1 MHz	3 s	1 s interval	few hours - few months	4 GB
E	1024 S/C (V) 256 S/C (I) (x 50/60 Hz)	+ 1 year	1 min - 60 min	$\approx$ 2.5 GB/Month	32 GB
F	256 S/C up to 5 MHz	1 h	10 min	30 days	2 GB (SD card)
G	20 kHz, 200 kHz up to 2 MHz	200 ms	1 s; 2h interval	11.5 days; 55 weeks	2 - 32 GB (SD card)
H	6.4/7.68 kHz (128 S/C $\times$ 50/60 Hz)	150 ms - 40 s	10/12 Hz; 50/60 Hz	45 days	16 Mb (DRAM) 80 GB (with a PC)
I	16 kHz	1 - 90 s	1/2 cyc interval	90 min	16 GB (SDHC)

all the eight channels, it is possible to store 11 days of continuous waveform recording.

D - Sampling frequency of 512 S/C in 50/60 Hz, and 64 S/C in 400 Hz. Up to 1 MHz of sampling rate, for high speed TA. A continuous waveform register is updated every three seconds. The equipment logs the RMS and power values once per second. No waveforms are recorded under this monitoring mode. The monitoring time length can vary from few hours to few months or more depending on the user application.

E - Sampling frequency up to 1024 S/C for voltage monitoring and 256 S/C for current monitoring, or 512 S/C for both. More than one year of continuous waveform recording using a 32 GB mass memory, depending on the selected resolution and the grid condition.

F - Sampling frequency of 256 S/C synchronized with a range from 42.5 to 69 Hz. An ultra-high sampling rate of 5 MHz for TA is available. The device is also able to perform one hour of continuous waveform recordings and 30 days of parameter recordings using a 2 GB external SD card memory.

G - Sampling frequency of 20 kHz for waveform recordings. For parameter recordings it uses a 200 kHz sampling rate, and up to 2 MHz for TA applications. A parameter recording length of 11.5 days is possible setting 1 s of time plot interval and 55 weeks with 2 h interval.

H - Waveform recording length of up to 40 s with a sampling rate of 128 S/C, and 45 days of data recording with a sampling rate of one fifth of the power frequency. It requires an industrial PC to expand the memory capacity.

I - With a 16 kHz sampling frequency it is possible to perform up to 90 s of continuous waveform recording. A 90 min length of parameter recording is possible with a configurable interval between 1 and 3000 periods of the power frequency, which is tracked by the device. The mass memory is a 16 GB SDHC card.

Observing Table I, and taking into account the salient points of each manufacturer, it is possible to notice that all of them are able to record the PQ parameters for a long time period. Depending on the aggregation time, some equipment are able to record up to tens of weeks of PQ parameters. Nevertheless, only two of them are able to save the waveform recordings for a long period, the equipment of C and E brands. While C is able to record a couple of days using a huge amount of memory, E is able to register over than a year of waveform recordings due its compression process [20].

From this, it can be noticed that there is a shortage of devices that do the continuous waveform recording, currently. And above all, a suitable compression technique can provide a more efficient usage of mass memory, allowing longer recording lengths, and no losing of valuable signal information (including disturbances).

Although it is important to save all the information needed for any future oscillographic analysis or application of Digital Signal Processing (DSP) algorithms, it is important to do it using the least amount of memory. A way to deal with this balance is by using the Novelty Detection Concept. This concept avoid saving redundant information. For example, there is no sense to save many data that has the same information, as the stead state of the voltage or current signal, even if they have harmonic and interharmonic content.

In the Novelty Concept, the detecting process acts as a compression stage, selecting only the important parts of the signal that allows the whole waveform reconstruction.

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III. A NEW GAPLESS NOVELTY DETECTION RECORDING

Fig. 1 shows a general gapless waveform recorder in the new context of SG. The voltage and current signals from the grid are constantly acquired and monitored by the detector, which uses the Novelty Detection Concept [26] to identify novelties present in the signal. It allows storing a long time waveform recordings using lower amount of memory. The waveform recordings can be easily transported and used for many applications involving oscillographic analysis.

The Novelty Detection Concept addresses the detection of any abnormal variation in the signal waveform, not only the known disturbances, being suitable for application in the SGs scenario, where the type of the disturbances are not necessarily known. Based on this concept, the data storage is performed in a more efficient way, since only the pieces of signal that present novelties are sufficient to describe the signal behavior. To represent the signal between novelties, some auxiliary parameters can be stored, such as: the time interval or the number of cycles and the frequency of the signal. This approach reduces significantly the amount of data to be stored.

The proposed Novelty Detection Concept applied to a gapless waveform recorder of EPS signals lies in the fact that the signal from the power grid can be segmented in frames and the aim is to identify which frames are needed to be stored. It is done by detecting if a frame is different from a reference one. Thus, the application is as follows: (i) the signal is segmented in fixed length frames; (ii) the current frame is compared to a reference one; and (iii) if they were different, a novelty is signalized by the Novelty Flag. This process is shown in Fig. 2.

A. Proposal for Implementation of a Novelty Detector

The first step for the implementation of a general novelty detector is the signal segmentation, whose length can be fixed or variable (adaptive segmentation), depending on the signal behavior and the processing technique.

The aim of the adaptive segmentation is to find the segment length where the signal can be represented by the same model parameters. It has been extensively used in the biomedical and speech signal processing [27]. A similar concept has been introduced in [28] to analyze power system disturbances. The drawback of the adaptive segmentation is that it needs extra processing to identify the optimal length for each segment.

Fig. 3 shows a representation of a normalized 60 Hz power signal which was segmented into fixed length frames with four fundamental component cycles to each frame. The frames boundaries are shown in dashed line. The signal contains a high frequency transient disturbance, typically caused by a capacitor switching, that leads to a damped high frequency oscillation. In the figure, a transient begins at about $t = 0.12$ s. A total of five frames are shown. It can be seen that the disturbance occurs between frames 2 and 3. In this way, at least frames 1 to 4, which are highlighted in the shaded area, must be stored to preserve all the signal behavior, including the non-stationarity. So, these frames should be flagged as novelty frames. After the frame 4, the signal is a "copy" of it, as long as it remains stationary.

In Fig. 4, it can be seen that frames of one fundamental cycle length is more suitable for the given disturbance. From this segmenting process, frames 1 and 8 to 10 must be stored in order to recovery the signal. After the tenth frame the signal is a "copy" of it. Observing both Fig. 3 and Fig. 4, it is clear that, for this example, the latter shows a more efficient segmentation process than that one shown in the former, since less information is stored and the same signal can be reconstructed. Therefore, the effectiveness of compression is highly related to how the signal is segmented, and an appropriate segmentation depends on the type of the analyzed signal and disturbances.

There are several segmentation methods, some of them focused on adaptive segmentation, and others on finding out stationarity changing points in the signal. Among them it can be mentioned the monitoring of the Root Mean Square (RMS) value, the applying of digital high-pass filters, the detection of singular points from wavelet transforms, tracking of spectral changing, among others [28], [29].

After the segmentation, the next step is the comparison between the frames. This process is responsible to identify those important highlighted frames in Fig. 3 and Fig. 4. They are important because they have all the novelties present in the signal. This comparison is related to some metric and a threshold is necessary to determine if the compared frames are different or not. Many metrics can be used to perform the frame comparison, the most common for power systems signals are: Mean Squared Error (MSE), Cross Correlation, RMS value, Energy, Magnitude Spectrum (through Fas Fourier Transform - FFT), among others.

The thresholding process can be performed either with a fixed level or adaptively. Regarding the second type, the value of the threshold can be adapted according to the quantity of previous sequential detections, or even due to the noise scenario. For example, if there are too much detections sequentially, the threshold can be increased automatically, leading to a less sensitive detector and hence, fewer detections must be triggered. On the other hand, if there are very few detections, the threshold value can be decreased in order to get a more sensitive detector.

An example of a Novelty Detector application based on the energy comparison between frames is shown in Fig. 5. A more detailed view of the detector architecture, including the theoretical basis, mathematical formulation, compression/reconstruction algorithms and schemes can be found in [26] and [30].

The signal shown in the figure represents a portion of 1 hour length of the current signal from one phase of a wind generator. In this figure, the signal frames that present novelties are plotted in black, and the frames where there were no novelties are plotted in gray. It was chosen frames with a length of four cycles of the fundamental component and a threshold in the energy difference between frames was applied.

It can be noted that this approach reduces the amount of the

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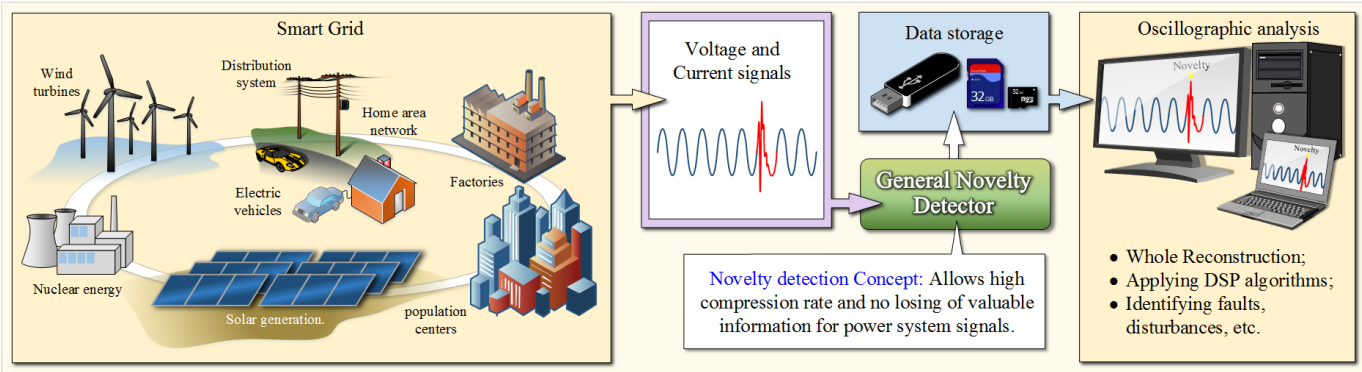

Fig. 1: Gapless waveform recording approach into Smart Grid.

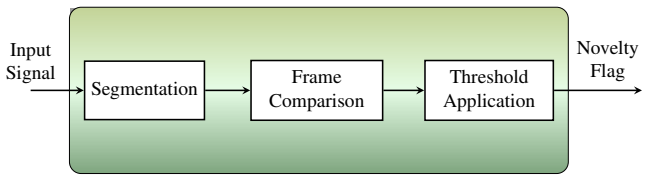

Fig. 2: Block diagram of a General Novelty Detector.

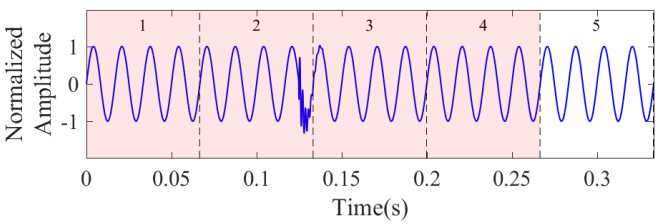

Fig. 3: Segmenting with a frame lengh of four fundamental component cycles.

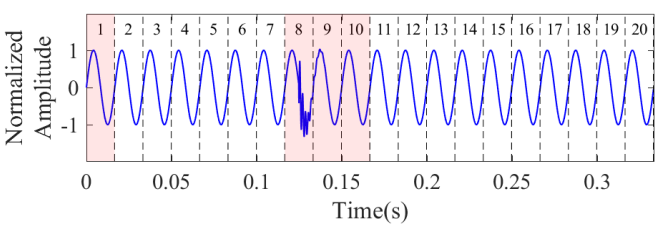

Fig. 4: Segmenting with one fundamental component cycle frame length.

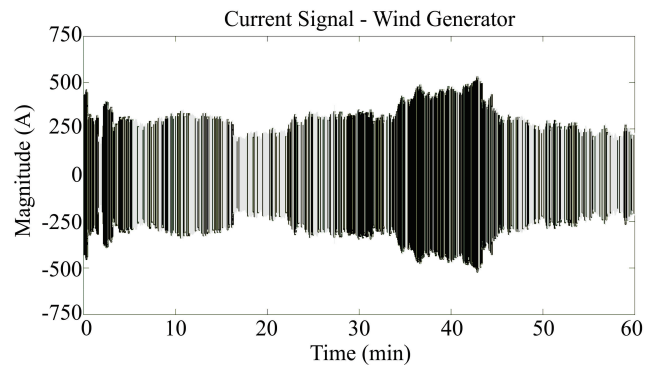

Fig. 5: Current signal from a wind generator.

stored data, since only the novelty frames were fully stored. The total duration of the signal is approximately 34 hours. Considering a sampling rate of 128 S/C, 16 bits per sample (2 B/S), and eight channels, it wold be necessary 15041 MB ($\approx$ 14.7 GB) of memory to store all the samples of the signal. By using the Novelty Detection Concept it was possible to spend only 27 MB of memory to store the signal. Thus, the compression rate was of 557:1.

Another example is shown in Fig. 6, which uses the same color scheme of Fig. 5. This signal is composed by a single component which is changed both in amplitude and in frequency every 20 s interval. The amplitude step is about 5 V, from 140 V to 210 V. Simultaneously, the frequency is changed with a 0.1 Hz step from 59 Hz to 61 Hz. Once all values reach the maximum, they all return to the original ones with the same behavior. With about 10 min length, it would be necessary 9.96 MB to store all the signal samples. With the use of the detector, the compression led to a 107 kB file. Although it is a synthetic signal, a remark that can be highlighted is about the fact that real voltage signals present less detections than current signals, since they suffer less variations in time due their smoother behavior. Hence, the compression rate can be several times higher for the same length.

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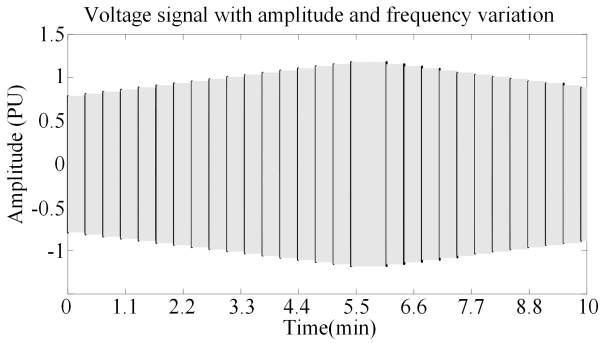

Fig. 6: Voltage signal with changes in amplitude and frequency.

Obviously, with the use of the Novelty Detection Concept, some parts of the signal are not saved, but as they are redundant, the whole signal can be reconstructed. The losing of information can be controlled by an appropriate choice of the threshold. Thus, if the total storing of the signal is wanted, it is enough to set the threshold to save all frames. In this condition, the operation of the detector is the same as traditional waveform recorders.

IV. Conclusion

This paper presented the Novelty Detection Concept and its application in Electric Power System signals. This approach is necessary in Smart Grids (SGs) context, since events not fully known may occur. It is based on segmenting the signals in frames and identifying which frames have novelties and must to be saved.

The application of this concept proved to be very effective in storing long periods of waveforms signals even those which contain a large amount of disturbances such as current signals. Despite the high compression ratio, it was possible to perform the complete reconstruction of the waveform signal, because the Novelty Concept preserves all the important parts of the signal and discard the redundant parts without affect the integrity of the signal.

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